

The Zero Mellin Mode of a Prime-Target Residual: Fixed-Shift Titchmarsh Closure, a Large-Factor Subtraction Formula, and the Scale-Localization Boundary

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Abstract

Let $h \neq 0$ be fixed and let $W \in C_c^1((0, \infty))$. We study the prime-target residual aggregated over all factor scales,

$$\mathcal{A}_h(X; W) = \sum_{m, n \geq 1} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X),$$

and its large-factor portion $\mathcal{A}_{h, > D}$ obtained by imposing $n > D$. Although $\Lambda - 1$ has L^2 energy of order $N \log N$ on a dyadic interval, the complete factor aggregation has the exact collapse

$$\sum_{n|r} (\Lambda(n) - 1) = \log r - \tau(r).$$

Consequently $\mathcal{A}_h$ is a difference of a shifted prime number theorem and a Titchmarsh divisor sum.

Write

$$C_h(s) = \prod_{p|h} (1 - p^{-s-1}) \prod_{p \nmid h} \left(1 + \frac{1}{(p-1)p^{s+1}} \right), \quad C_h = C_h(0).$$

Using the uniform Titchmarsh theorem of Assing–Blomer–Li, we obtain a fixed- h asymptotic with arbitrary logarithmic saving. Classical maximal Bombieri–Vinogradov then evaluates the whole range $n \leq D$, uniformly for

$$D \leq X^{1/2} (\log X)^{-B}.$$

Subtracting gives an explicit asymptotic for the complementary range $n > D$, even though most of that range lies beyond the direct arithmetic-progression reduction. When $D = D(X) \rightarrow \infty$, its logarithmic leading term is

$$(1 - C_h) X \log(X/D) \int_0^\infty W(u) du.$$

Whenever $(1 - C_h)I_0(W) \neq 0$, its $X \log X$ coefficient at the square-root cutoff is one half of the complete-sum coefficient. For $h = 2$,

$$C_2 = \frac{1}{3} \frac{\zeta(2)\zeta(3)}{\zeta(6)} = 0.647865\dots,$$

so the coefficient is explicitly nonzero.

The result is a scale-aggregation theorem, not a new Titchmarsh, Bombieri–Vinogradov, dispersion, or Type II estimate. The factor collapse is the zero Mellin mode of the multiplicative scale variable. Nonzero Mellin modes do not reduce to $\log r - \tau(r)$, and a single factor layer already contains fixed-shift prime-pair covariance. In particular, the paper proves no fixed prime-pair asymptotic, twin-prime lower bound, or breach of the parity barrier.

Keywords: prime-target residual; Titchmarsh divisor problem; Bombieri–Vinogradov theorem; factor-scale aggregation; Mellin localization; sieve parity.

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1 Introduction and exact scope

The preceding finite-conductor transfer result separates a bounded periodic second coefficient from the energy-bearing remainder of $\Lambda - 1$. The periodic part can be passed through $\Lambda(mn + h)$ in a one-sided Bombieri–Vinogradov range, but low-period spaces capture only a vanishing fraction of the L^2 energy of the prime fluctuation [12, 13]. The next question is therefore not whether that remainder is large—it is—but whether a specified prime-target form can nevertheless average it.

This paper answers that question in one nonlocal coordinate. Fix $h \in \mathbb{Z} \setminus \{0\}$ and $W \in C_c^1((0, \infty))$. We extend $\Lambda(t)$ by zero for $t \leq 0$ and define

$$\mathcal{A}_h(X; W) = \sum_{m, n \geq 1} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X). \quad (1)$$

For $D \geq 1$, put

$$\mathcal{A}_{h, \leq D}(X; W) = \sum_{\substack{m, n \geq 1 \\ n \leq D}} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X), \quad (2)$$

$$\mathcal{A}_{h, > D}(X; W) = \sum_{\substack{m, n \geq 1 \\ n > D}} (\Lambda(n) - 1) \Lambda(mn + h) W(mn/X). \quad (3)$$

The split is exact:

$$\mathcal{A}_h = \mathcal{A}_{h, \leq D} + \mathcal{A}_{h, > D}. \quad (4)$$

The form (1) does not localize m and n separately. It fixes their product scale and sums over every factorization. That distinction is the source of the theorem. If $r = mn$, then

$$\sum_{n|r} (\Lambda(n) - 1) = \log r - \tau(r). \quad (5)$$

Thus the full factor coordinate collapses to the difference between an elementary logarithmic weight and the divisor function. The target remains prime-sensitive, but it is now precisely a Titchmarsh divisor problem.

1.1 The fixed-shift result

Set

$$I_0(W) = \int_0^\infty W(u) du, \quad I_1(W) = \int_0^\infty W(u) \log u du, \quad (6)$$

and define the Euler product

$$C_h(s) = \prod_{p|h} (1 - p^{-s-1}) \prod_{p \nmid h} \left(1 + \frac{1}{(p-1)p^{s+1}} \right). \quad (7)$$

It is holomorphic in a neighborhood of $s = 0$. We abbreviate

$$C_h = C_h(0), \quad C'_h = C'_h(0). \quad (8)$$

Here and below, γ denotes the Euler–Mascheroni constant.

Our main theorem, stated precisely in [theorem 4.1](#), gives

$$\begin{aligned} \mathcal{A}_{h,>D}(X; W) = & X \left[(1 - C_h) \{ (\log X) I_0(W) + I_1(W) \} - 2(\gamma C_h + C'_h) I_0(W) \right] \\ & - X I_0(W) \sum_{\substack{n \leq D \\ (n,h)=1}} \frac{\Lambda(n) - 1}{\varphi(n)} + O_{A,h,W} \left(\frac{X}{(\log X)^A} \right), \end{aligned} \quad (9)$$

uniformly for

$$2 \leq D \leq X^{1/2} (\log X)^{-B}, \quad (10)$$

where $B = B(A, h, W)$ is sufficiently large. An elementary Dirichlet-series calculation then shows

$$\sum_{\substack{n \leq D \\ (n,h)=1}} \frac{\Lambda(n) - 1}{\varphi(n)} = (1 - C_h) \log D + \kappa_h + o_h(1) \quad (11)$$

for an explicit finite constant κ_h . Hence the leading tail is

$$(1 - C_h) X I_0(W) \log(X/D). \quad (12)$$

This is a fixed-shift theorem, including $h = 2$. It evaluates the complete range beyond the direct Bombieri–Vinogradov cutoff, but only after all of those factor scales have been added with their original signs. No estimate for an individual dyadic n -layer follows.

1.2 What is new and what is imported

The analytic estimates used here are established results. The complete sum is evaluated by the uniform Titchmarsh theorem of Assing et al. [1], whose work supplies strong errors and substantial uniformity in the shift. Earlier decisive forms of the Titchmarsh divisor problem include work of Fouvry [5] and Bombieri et al. [3]. The range $n \leq D$ is evaluated by the classical maximal Bombieri–Vinogradov theorem [2, 11]. We prove no improvement to either input.

The contribution is instead the exact residual coordinate and its boundary:

- (i) the full prime fluctuation $\Lambda(n) - 1$, not a bounded periodic approximation, is inserted into a genuine fixed-shift prime target;
- (ii) complete factor aggregation converts it exactly into $\log r - \tau(r)$;
- (iii) the small-factor Bombieri–Vinogradov evaluation is subtracted from the complete Titchmarsh evaluation, yielding the entire large-factor tail;
- (iv) Mellin twisting identifies why only the zero scale frequency collapses and why factor localization reintroduces the unresolved bilinear geometry.

This distinction parallels a general lesson of the asymptotic sieve: distribution information and the bilinear information needed to overcome parity are separate inputs [6]. Recent work on prime-producing sieves makes the Type I/Type II ledger still more explicit [4]. Our theorem resolves one aggregated coordinate; it does not verify the bilinear hypotheses required in those frameworks.

For comparison, prime-pair asymptotics are known for almost all shifts in long windows, with the strongest two-point range relevant here due to Matomäki et al. [9]. Newer higher-uniformity results handle arbitrary fixed one-parameter prime-shift patterns $n + a_i h$ in an almost-all- h regime [10]. Neither statement selects a prescribed fixed shift such as $h = 2$.

1.3 Strict claim boundary

The following statements are not consequences of this paper.

- B1.** We do not prove an asymptotic for $\sum_n \Lambda(n)\Lambda(n+h)W(n/X)$ at any prescribed admissible even shift h ; in particular, we do not treat $h = 2$.
- B2.** We do not estimate a fixed dyadic box $m \asymp M, n \asymp N$ with two unrestricted coefficients.
- B3.** The large-factor theorem permits cancellation between different factor scales. It supplies neither pointwise-in-scale control nor an L^2 estimate in the scale variable.
- B4.** No new level of distribution, dispersion theorem, Type II estimate, twin-prime lower bound, or breach of the sieve parity barrier is asserted.
- B5.** The phrase “zero Mellin mode” refers to multiplicative factor-scale aggregation. It is not a spectral realization of the primes or an isomorphism with a nonlinear map.

The value of the result is precisely that both sides of the boundary are rigorous: the zero scale frequency closes at every fixed h , while the localization step that would isolate a prime-pair layer is displayed rather than assumed.

2 Complete factor aggregation and Titchmarsh closure

We first evaluate the complete sum (1). The algebraic reduction is elementary, but it is the point at which the factor-scale geometry changes.

Proposition 2.1 (Exact zero-mode collapse). *For every $X > 0$,*

$$\mathcal{A}_h(X; W) = \sum_{r \geq 1} (\log r - \tau(r)) \Lambda(r+h) W(r/X). \quad (13)$$

Proof. Only finitely many pairs (m, n) occur because W is compactly supported. Group them by $r = mn$. The coefficient of $\Lambda(r+h)W(r/X)$ becomes

$$\sum_{n|r} (\Lambda(n) - 1) = \sum_{n|r} \Lambda(n) - \sum_{n|r} 1.$$

The convolution identities

$$\Lambda * \mathbf{1} = \log, \quad \mathbf{1} * \mathbf{1} = \tau$$

give (13). □

2.1 The two analytic inputs

We record the smooth fixed-shift forms needed below. The first is an elementary consequence of the prime number theorem; the second follows from the uniform Titchmarsh divisor theorem of Assing et al. [1].

Lemma 2.2 (Shifted logarithmic prime weight). *For every $A > 0$,*

$$\sum_{r \geq 1} \log r \Lambda(r+h) W(r/X) = X \{ (\log X) I_0(W) + I_1(W) \} + O_{A,h,W} \left(\frac{X}{(\log X)^A} \right). \quad (14)$$

Proof. Put $t = r + h$ and apply the prime number theorem with its classical zero-free-region error to the Stieltjes integral against

$$F_X(t) = \log(t-h) W((t-h)/X).$$

For sufficiently large X , the support stays in the positive t -axis. The continuous main term is

$$\int_0^\infty \log r W(r/X) dr = X \{(\log X)I_0(W) + I_1(W)\}.$$

The derivative of F_X costs only a fixed power of $\log X$, which is absorbed by the zero-free-region saving. $\square$

Theorem 2.3 (Smooth uniform Titchmarsh input). *For every $A > 0$,*

$$\begin{aligned} \sum_{r \geq 1} \tau(r) \Lambda(r+h) W(r/X) &= X C_h \{(\log X)I_0(W) + I_1(W) + 2\gamma I_0(W)\} \\ &\quad + 2X C'_h I_0(W) + O_{A,h,W} \left(\frac{X}{(\log X)^A} \right). \end{aligned} \quad (15)$$

Proof. Apply Assing et al. [1, Theorem 1.1] with prime variable $t = r + h$, sign parameter $\sigma = 1$, and shift $f = -h$. Their Euler product $c_s(f)$ is exactly $C_h(s)$ in (7), since it depends on the prime divisors of f only. To check the endpoint hypotheses, enclose $\text{supp } W$ in $[a, b] \subset (0, \infty)$. On every sharp subinterval used in partial summation, let $Y < X_1$ denote the corresponding endpoints in the prime variable t . Since $t = r + h \asymp X$ and h is fixed, the conditions $f \neq 0$, $\max(-f, 0) \leq Y < X_1$, and $X_1 + f \asymp X_1$ hold once X is sufficiently large. Their sharp-interval main term has derivative, in the r variable,

$$C_h \log r + 2\gamma C_h + 2C'_h.$$

Stieltjes partial summation against $W(r/X)$ therefore gives (15). The cited theorem supplies an arbitrary power of $\log X$ in the error, so the fixed derivatives of W are harmless. $\square$

Remark 2.4 (Provenance). The deep analytic content of [theorem 2.3](#) belongs to Assing–Blomer–Li. We include the deduction to fix the shift convention, the derivative term $2C'_h$, and the smoothing normalization; no new Titchmarsh estimate is claimed.

2.2 The complete residual asymptotic

Theorem 2.5 (Fixed-shift zero-Mellin closure). *Fix $h \neq 0$ and $W \in C_c^1((0, \infty))$. For every $A > 0$,*

$$\begin{aligned} \mathcal{A}_h(X; W) &= X \left[(1 - C_h) \{(\log X)I_0(W) + I_1(W)\} \right. \\ &\quad \left. - 2(\gamma C_h + C'_h)I_0(W) \right] + O_{A,h,W} \left(\frac{X}{(\log X)^A} \right). \end{aligned} \quad (16)$$

Proof. Insert (14) and (15) into the exact identity (13) and subtract. $\square$

The coefficient $1 - C_h$ is not introduced by a heuristic independence model. It is the exact difference between the logarithmic divisor mass and the leading Titchmarsh density in the completely aggregated coordinate.

Corollary 2.6 (The shift two constant). *For $h = 2$,*

$$C_2 = \frac{1}{3} \frac{\zeta(2)\zeta(3)}{\zeta(6)} = 0.647865\dots, \quad (17)$$

so $1 - C_2 > 0$.

Proof. At $s = 0$,

$$C_2 = \frac{1}{2} \prod_{p>2} \left(1 + \frac{1}{p(p-1)}\right).$$

The local identity

$$1 + \frac{1}{p(p-1)} = \frac{1 - p^{-6}}{(1 - p^{-2})(1 - p^{-3})}$$

gives

$$\prod_p \left(1 + \frac{1}{p(p-1)}\right) = \frac{\zeta(2)\zeta(3)}{\zeta(6)}.$$

Removing the $p = 2$ factor $3/2$ proves the exact formula. The numerical value is included only for orientation. $\square$

Remark 2.7 (Energy versus aggregated influence). On $(N, 2N]$ one has $\|\Lambda - 1\|_2^2 \sim N \log N$. Thus [theorem 2.5](#) does not come from a small source coefficient. It shows instead that a large-energy coefficient can have a rigorously computable prime-target influence after a sufficiently nonlocal aggregation. This is compatible with, and sharper in a different norm than, the finite-conductor energy escape in TPC-7.

3 The small-factor transfer

We next evaluate [\(2\)](#). Here n is the modulus seen by the target prime. This is a one-sided Type I problem, so the classical square-root level applies.

Theorem 3.1 (Coefficient-uniform fixed-product Type I estimate). *Fix $h \neq 0$, $W \in C_c^1((0, \infty))$, and $A > 0$. There is $B = B(A, h, W) > 0$ such that, whenever*

$$1 \leq D \leq \frac{X^{1/2}}{(\log X)^B}, \tag{18}$$

one has, uniformly for arbitrary coefficients $|a_n| \leq 1$,

$$\sum_{\substack{n \leq D \\ (n, h) = 1}} a_n \left\{ \sum_{m \geq 1} \Lambda(nm + h) W(nm/X) - \frac{X}{\varphi(n)} I_0(W) \right\} \ll_{A, h, W} \frac{X}{(\log X)^A}. \tag{19}$$

Proof. For $(n, h) = 1$, the substitution $t = nm + h$ places the target in the reduced class $h \pmod{n}$ and changes the weight to $W((t - h)/X)$. Stieltjes partial summation bounds the row error by a constant depending on W times

$$\max_{y \ll_W X + |h|} \left| \sum_{\substack{t \leq y \\ t \equiv h \pmod{n}}} \Lambda(t) - \frac{y}{\varphi(n)} \right|.$$

The continuous main term is exactly

$$\frac{1}{\varphi(n)} \int_0^\infty W((t - h)/X) dt = \frac{X}{\varphi(n)} I_0(W).$$

Sum the maximal row errors over $n \leq D$ and apply the classical maximal Bombieri–Vinogradov theorem with a slightly larger requested logarithmic saving [\[2, 8, 11\]](#). $\square$

Rows not coprime to h are supported on a finite collection of prime-power targets. We isolate this elementary point because the source coefficient is now $\Lambda(n) - 1$, rather than a Möbius divisor weight.

Lemma 3.2 (Noncoprime residual rows). *For every $D \leq 2X$ and every $\varepsilon > 0$,*

$$\sum_{\substack{n \leq D \\ (n, h) > 1}} |\Lambda(n) - 1| \sum_{m \geq 1} \Lambda(nm + h) |W(nm/X)| \ll_{h, W, \varepsilon} X^\varepsilon. \quad (20)$$

Without an artificial upper cutoff, one also has

$$\sum_{\substack{m, n \geq 1 \\ n > D \\ (n, h) > 1}} |\Lambda(n) - 1| \Lambda(nm + h) |W(nm/X)| \ll_{h, W, \varepsilon} X^\varepsilon. \quad (21)$$

Proof. For either displayed sum, put $r = nm$ and enlarge the divisor range. The relevant left-hand side is at most

$$\sum_{r \asymp_W X} \Lambda(r + h) |W(r/X)| \sum_{\substack{n | r \\ (n, h) > 1}} (\Lambda(n) + 1).$$

If the inner sum is nonempty, some prime $p \mid h$ divides r . Then $p \mid r + h$. If $\Lambda(r + h) \neq 0$, it follows that $r + h = p^k$. For each of the finitely many primes dividing h , only $O_{h, W}(1)$ powers lie in the fixed-ratio interval forced by the support of W . Finally,

$$\sum_{n | r} (\Lambda(n) + 1) = \log r + \tau(r) \ll_\varepsilon X^\varepsilon,$$

after reducing ε to absorb the remaining logarithms. $\square$

Theorem 3.3 (Small-factor prime-residual transfer). *Fix $A > 0$ and choose $B = B(A, h, W)$ sufficiently large. Uniformly for*

$$2 \leq D \leq \frac{X^{1/2}}{(\log X)^B}, \quad (22)$$

one has

$$\mathcal{A}_{h, \leq D}(X; W) = X I_0(W) \sum_{\substack{n \leq D \\ (n, h) = 1}} \frac{\Lambda(n) - 1}{\varphi(n)} + O_{A, h, W} \left(\frac{X}{(\log X)^A} \right). \quad (23)$$

Proof. The noncoprime rows are absorbed by [lemma 3.2](#). On the remaining rows apply [theorem 3.1](#) with

$$a_n = \frac{\Lambda(n) - 1}{1 + \log D}.$$

These coefficients have modulus at most one. Multiply the result back by $1 + \log D$ and request $A + 3$ logarithmic saving in the Type I theorem. This proves [\(23\)](#). $\square$

Remark 3.4 (Why the coefficient is permitted here). The source factor $\Lambda(n) - 1$ is unbounded and nonperiodic, but in [theorem 3.3](#) it weights the *modulus* average. After division by $1 + \log D$, maximal Bombieri–Vinogradov is uniform in that row coefficient. This does not create a free second coefficient inside an individual prime progression and therefore does not constitute a Type II estimate.

4 The large-factor tail and the scale-localization boundary

We now subtract the direct Type I range from the complete zero-mode asymptotic. This evaluates an aggregate lying mostly beyond the Bombieri–Vinogradov cutoff.

Theorem 4.1 (Complete large-factor residual tail). *Fix $h \neq 0$, $W \in C_c^1((0, \infty))$, and $A > 0$. There is $B = B(A, h, W) > 0$ such that, uniformly for*

$$2 \leq D \leq \frac{X^{1/2}}{(\log X)^B}, \quad (24)$$

one has

$$\begin{aligned} \mathcal{A}_{h, > D}(X; W) = & X \left[(1 - C_h) \{ (\log X) I_0(W) + I_1(W) \} - 2(\gamma C_h + C'_h) I_0(W) \right] \\ & - X I_0(W) \sum_{\substack{n \leq D \\ (n, h)=1}} \frac{\Lambda(n) - 1}{\varphi(n)} + O_{A, h, W} \left(\frac{X}{(\log X)^A} \right). \end{aligned} \quad (25)$$

Proof. Use the exact split (4). Substitute the complete evaluation (16) and the small-factor evaluation (23), then subtract. $\square$

The theorem keeps the finite coefficient sum visible because this gives an arbitrary logarithmic error uniformly throughout the stated range. Its asymptotic shape follows from a Dirichlet-series calculation.

Lemma 4.2 (The residual coefficient average). *There is a finite constant κ_h such that*

$$\sum_{\substack{n \leq D \\ (n, h)=1}} \frac{\Lambda(n) - 1}{\varphi(n)} = (1 - C_h) \log D + \kappa_h + o_h(1) \quad (D \rightarrow \infty). \quad (26)$$

More precisely,

$$\sum_{\substack{n \leq D \\ (n, h)=1}} \frac{1}{\varphi(n)} = C_h \log D + \gamma C_h + C'_h + o_h(1), \quad (27)$$

and

$$\sum_{\substack{n \leq D \\ (n, h)=1}} \frac{\Lambda(n)}{\varphi(n)} = \log D + \kappa_{\Lambda, h} + o_h(1), \quad (28)$$

where

$$\kappa_h = \kappa_{\Lambda, h} - \gamma C_h - C'_h. \quad (29)$$

One may take

$$\kappa_{\Lambda, h} = -\gamma + \sum_{p \nmid h} \frac{\log p}{(p-1)^2} - \sum_{p \mid h} \frac{\log p}{p-1}, \quad (30)$$

whose infinite sum is absolutely convergent.

Proof. For $\Re s > 0$, an Euler-factor calculation gives

$$\sum_{\substack{n \geq 1 \\ (n, h)=1}} \frac{1}{\varphi(n) n^s} = \zeta(s+1) C_h(s). \quad (31)$$

Indeed, for $p \nmid h$ the Euler factor left after removing the $\zeta(s+1)$ factor is

$$1 + \frac{1}{(p-1)p^{s+1}},$$

whereas for $p \mid h$ it is $1 - p^{-s-1}$. The product $C_h(s)$ is absolutely convergent in a neighborhood to the right of $\Re s = -1/2$. Convolving its coefficients with $1/n$, and using $\sum_{k \leq y} k^{-1} = \log y + \gamma + o(1)$, gives (27); the constant term is $\gamma C_h + C'_h$.

For the von Mangoldt numerator,

$$\sum_{\substack{n \geq 1 \\ (n, h) = 1}} \frac{\Lambda(n)}{\varphi(n)n^s} = \sum_{p \nmid h} \frac{(\log p)p^{-s}}{(p-1)(1-p^{-s-1})}. \quad (32)$$

An explicit prime-power comparison avoids any Tauberian issue. Namely,

$$\sum_{\substack{n \leq D \\ (n, h) = 1}} \frac{\Lambda(n)}{\varphi(n)} = \sum_{n \leq D} \frac{\Lambda(n)}{n} + \sum_{\substack{p \nmid h, k \geq 1 \\ p^k \leq D}} \frac{\log p}{p^k(p-1)} - \sum_{\substack{p \mid h, k \geq 1 \\ p^k \leq D}} \frac{\log p}{p^k}.$$

The last two sums converge absolutely as $D \rightarrow \infty$, while the standard prime number theorem consequence $\sum_{n \leq D} \Lambda(n)/n = \log D - \gamma + o(1)$ gives (28) with (30). Subtracting (27) proves (26) and (29). $\square$

Corollary 4.3 (Asymptotic shape of the tail). *Suppose $D = D(X) \rightarrow \infty$ and that (24) holds. Then*

$$\begin{aligned} \mathcal{A}_{h, > D}(X; W) &= (1 - C_h)X I_0(W) \log(X/D) \\ &\quad + X \left[(1 - C_h)I_1(W) - \{2(\gamma C_h + C'_h) + \kappa_h\}I_0(W) \right] + o_{h, W}(X). \end{aligned} \quad (33)$$

Proof. Insert (26) into (25). Choose one fixed $A > 1$; the displayed error is then $o(X)$. $\square$

Corollary 4.4 (Half of the leading logarithmic coefficient). *Let*

$$D = \left\lfloor \frac{X^{1/2}}{(\log X)^B} \right\rfloor \quad (34)$$

with fixed sufficiently large B . If $(1 - C_h)I_0(W) \neq 0$, then

$$\mathcal{A}_{h, > D}(X; W) = \frac{1 - C_h}{2} X I_0(W) \log X + O_{h, W, B}(X \log \log X). \quad (35)$$

Proof. Use

$$\log(X/D) = \frac{1}{2} \log X + B \log \log X + o(1)$$

in (33). $\square$

Remark 4.5 (What “beyond Bombieri–Vinogradov” means). The moduli $n > D$ in theorem 4.1 are not controlled one by one or in dyadic blocks by the proof. Their *complete signed sum* is known because it equals the Titchmarsh-evaluated total minus the Bombieri–Vinogradov-evaluated short part. The theorem must not be read as a new level of distribution for those moduli.

4.1 Why the collapse is only the zero Mellin mode

The factorization $r = mn$ carries a multiplicative scale coordinate. To see what is lost by complete aggregation, twist the divisor pair by $(n/m)^{it}$. For every real t ,

$$\sum_{mn=r} (\Lambda(n) - 1)(n/m)^{it} = r^{-it} \left\{ \sum_{d \mid r} \Lambda(d) d^{2it} - \sigma_{2it}(r) \right\}. \quad (36)$$

At $t = 0$ this is $\log r - \tau(r)$. For $t \neq 0$, the first divisor sum no longer collapses to $\log r$, and the second is a genuinely twisted divisor function; here $\sigma_z(r) = \sum_{d|r} d^z$.

If V is a smooth cutoff in the ratio n/m , Mellin inversion writes

$$\widehat{V}(t) = \int_0^\infty V(y) y^{-it} \frac{dy}{y}, \quad (37)$$

and, with this convention,

$$V(n/m) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{V}(t) (n/m)^{it} dt \quad (38)$$

for a smooth compactly supported V . Thus localizing a factor box requires a continuum of nonzero frequencies. The value at $t = 0$ does not control that integral, just as the mean of a function does not control its pointwise values.

4.2 The fixed layer already contains the prime-pair problem

The $m = 1$ layer of the complete form is

$$\sum_{n \geq 1} (\Lambda(n) - 1) \Lambda(n + h) W(n/X). \quad (39)$$

The prime number theorem evaluates the part with coefficient -1 . An asymptotic for the remaining product $\Lambda(n) \Lambda(n + h)$ at a fixed admissible h is the Hardy–Littlewood prime-pair problem [7]. At $h = 2$ it is the twin-prime correlation. For the square-root choice of D , the full layer (39) lies inside $\mathcal{A}_{h, > D}$, but [theorem 4.1](#) allows it to cancel against all $m \geq 2$ layers. Therefore the aggregate theorem supplies no estimate for (39) by itself.

Adding a prime weight in the other factor does not preserve the collapse. Indeed,

$$\sum_{mn=r} \Lambda(m) (\Lambda(n) - 1) = (\Lambda * \Lambda)(r) - \log r. \quad (40)$$

The shifted target sum associated with $\Lambda * \Lambda$ is a parity-sensitive prime/semiprime correlation, not the Titchmarsh divisor sum in [theorem 2.3](#). This marks the next genuinely new analytic interface.

Remark 4.6 (A one-way implication ledger). The proved implications are

complete factor aggregation + Assing–Blomer–Li $\implies$ fixed- h zero-mode asymptotic,
maximal Bombieri–Vinogradov $\implies$ small-factor residual asymptotic,
the two evaluations and exact subtraction $\implies$ complete large-factor tail asymptotic.

None of these arrows reverses to give a dyadic Type II estimate or a fixed-layer prime-pair asymptotic.

5 Exact finite certificate and numerical orientation

The proof of the asymptotic theorem depends on the cited Titchmarsh and Bombieri–Vinogradov inputs, which no finite computation can certify. The algebraic layer, however, admits an exact machine-checkable certificate. We also record a separate floating-point diagnostic so that the two roles are not conflated.

5.1 Formal certificate

The program `experiments/exact_zero_mellin_certificate.py` represents each $\log p$ and each product $\log p \log q$ by a formal basis symbol with an integer coefficient. It therefore uses no floating approximation to the von Mangoldt function. With

$$1000 \leq r \leq 2000, \quad h = 2, \quad D = 31, \quad (41)$$

the committed certificate checks all of the following statements exactly:

- (a) for every one of the 1001 product values, $\sum_{d|r} (\Lambda(d) - 1) = \log r - \tau(r)$;
- (b) the direct sum over all 8465 factor pairs equals the collapsed product sum as a formal expression;
- (c) the 4035 pairs with $d \leq 31$ and the 4430 pairs with $d > 31$ recombine to the complete expression;
- (d) for every $1 \leq d \leq 31$, the substitution $t = dm + h$ agrees exactly with the corresponding arithmetic-progression indexing;
- (e) the local Euler factors at $s = 0$ used in (31) agree as rational numbers for every prime $p \leq 19$.

The resulting complete formal expression has 431 nonzero basis terms. The canonical payload SHA-256 is

6686C9E926CCB05F4DB0BF56A6A8F2811D5DED9A020FB905AC8DC54D350EDAEE,

and the SHA-256 of the indented committed JSON file is

A0D6303B26118D756406677A748D74160065390E5B4A55F70FDD2F1549E2BE43.

These hashes identify the finite conventions and artifact. They do not certify [theorems 2.3](#) and [3.1](#), nor are they used in their proofs.

5.2 A deliberately non-theorem diagnostic

For numerical orientation only, the second program replaces the smooth weight by the sharp box $X < r \leq 2X$, takes $h = 2$ and $D = \lfloor \sqrt{X} \rfloor$, and computes the exact decomposition using floating logarithms. The entries in [table 1](#) are normalized by $X \log X$. The last column is $X^{-1} \sum_{X < r \leq 2X} \Lambda(r + 2)$.

Table 1: Sharp-box implementation diagnostic; this is not a verification of the smooth asymptotic theorem.

X	D	complete	$n \leq D$	$n > D$	target mass
2^{14}	128	0.248166	0.039334	0.208832	0.996607
2^{16}	256	0.264715	0.054820	0.209896	1.001175
2^{18}	512	0.270705	0.071123	0.199581	1.001127
2^{20}	1024	0.277987	0.080975	0.197012	1.000196

For comparison, the predicted leading complete ratio is $1 - C_2 = 0.352134 \dots$, while the square-root tail leading ratio is $(1 - C_2)/2 = 0.176067 \dots$. The modest values of X , the discontinuous weight, lower-order terms, and the unspecified logarithmic loss in the admissible cutoff prevent this table from testing the theorem. Its purpose is narrower: it detects implementation mistakes in the complete/small/tail bookkeeping and displays the slow finite-scale drift.

Both layers are reproducible using only the Python standard library:

```
python experiments/exact_zero_mellin_certificate.py \
    --output experiments/data/exact-zero-mellin-certificate.json
python experiments/test_exact_zero_mellin_certificate.py
python experiments/zero_mellin_diagnostic.py \
    --powers 14,16,18,20 --h 2
python experiments/test_zero_mellin_diagnostic.py
```

6 Conclusion and the next analytic interface

The paper isolates one coordinate in which the full prime fluctuation can be passed through a prescribed fixed-shift prime target. Complete factor-scale aggregation turns $\Lambda - 1$ into $\log -\tau$ exactly. A proven Titchmarsh theorem evaluates the total, classical Bombieri–Vinogradov evaluates the short factor range, and subtraction evaluates the complete long-factor tail. When its leading coefficient is nonzero, the $X \log X$ coefficient at a square-root cutoff is one half of the complete-sum coefficient.

This positive result is deliberately paired with its limitation. It is a statement about one scalar scale coordinate, not about the distribution of the individual layers composing it. Mellin inversion shows that a dyadic factor localization requires nonzero scale frequencies, for which the elementary divisor collapse disappears. At the endpoint $m = 1$, the missing localized estimate is already the fixed-shift prime-pair problem.

The next defensible layer is therefore not to infer a localized theorem from the zero mode. It is to study the nonzero Mellin family in a region where existing dispersion or Type II estimates genuinely apply, keeping the coefficient class and averaging axes explicit. A successful result there would quantify how much scale resolution can be recovered. A negative result would still identify the exact norm or coefficient geometry that blocks recovery. Neither outcome is assumed here.

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